

Project Overview: Game of Thrones - Battle for Westeros

Repository: `redrighthand2007/Gameofthrones-Webgame`

Project Type: Strategy Web Game / Single Page Application (SPA)

Developer: `redrighthand2007`



The Elevator Pitch

A sleek, fast-paced strategy web game built entirely from scratch using **pure Vanilla Web Technologies** (Zero dependencies, no React, no NPM). Players act as military tacticians, drafting an alliance of iconic Game of Thrones houses within a strict 999G budget to battle an AI opponent.

⚡ Technology Stack & Architecture

- **Vanilla JavaScript (ES6+):** The entire game logic is modularized across three files:
 - `data.js`: State management, audio objects, and pricing matrix.
 - `ui.js`: DOM manipulation, navigation, and modal logic.
 - `combat.js`: AI generation, battle math, and inline DOM result injection.
- **HTML5:** Semantic layouts using `.hidden` class toggling to achieve instantaneous Single-Page Application (SPA) transitions without ever reloading the browser.
- **CSS3:** Advanced Flexbox and CSS Grid architecture, `overflow: hidden` UI locking, text-clipping gradients, hover-based modals, and custom particle animations.

✂ Core Gameplay Mechanics

1. **The Economy:** Both the Player and the AI have a strict budget of 999G.
2. **Drafting Strategy:** Houses are categorized by Role (Attack, Defense, Mix) and Tier (1 to 3). Prices scale rationally (e.g., Tier 1 Attack is 500G, Tier 3 Mix is 150G). Players select up to 3 units.
3. **The AI Opponent:** The computer simulates a master tactician by randomly selecting from one of three pre-programmed, mathematically flawless formations (Pure Attack, Pure Defense, or Pure Mix).
4. **Combat Resolution:** When "FIGHT!" is clicked, raw power is calculated and multiplied by a slight RNG factor (0.85x to 1.15x) to determine the fate of Westeros.

UI/UX Highlights

- **Cinematic Design:** A dark, atmospheric UI with glowing gold/crimson accents and a fixed background.
- **Dynamic Board:** The "War Room" uses a side-by-side flexbox layout. Selecting units instantly updates the player's horizontal roster and drains their real-time budget.
- **Inline Results:** When the battle concludes, the main title dynamically explodes into a glowing green "VICTORY" or blood-red "DEFEAT", spawning restart buttons directly inline to prevent jarring screen flashes.
- **Custom Audio Engine:** Features background music, sword clashes, and victory/defeat sound effects, all controllable via a custom top-right mute toggle overlay.
- **Hover Modals:** A pricing reference table dynamically appears via an invisible CSS/JS hover trigger over a settings gear icon.

What this project proves

This project is a masterclass in DOM manipulation, state management, and CSS layout architecture. It demonstrates the ability to build complex, state-driven interactive web applications without relying on heavy frontend frameworks like React or Vue.